SOLUTIONS: ESTIMATION AND FAULT DETECTION

1. Solution

a) According to the system state-space equations, the state matrix, the input matrix, and the output matrix of the system read as follows:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ K & g-K & 0 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}.$$

Thus, the observability matrix of the pair (C,A) is as follows:

$$O = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since O is not full rank, the system is not observable. As the non-observable subspace coincides with the kernel of O, a basis for the non-observable subspace is given by the vectors $[0 \;\; 1 \;\; 0]^\top$ and $[0 \;\; 0 \;\; 1]^\top$.

[5 Marks]

b) For any initial condition in the kernel of O, we have $y(t) = \theta_m(t) = 0$ for all time. Thus, any initial condition of the form $(\theta_m(0), \theta_\ell(0), \omega_\ell(0)) = (0, a, b)$ with $a, b \in \mathbb{R}$ produces zero output.

[3 Marks]

c) A basis for the orthogonal complement of the kernel of O is given by the vector $[1 \;\; 0 \;\; 0]^\top$. Therefore, the similarity transformation matrix T can be chosen by taking

$$T^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

so that

$$T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Therefore, the new state matrix, the input matrix, and the output matrix can be obtained as follows:

$$\bar{A} = TAT^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ g-K & 0 & K \\ 0 & 0 & 0 \end{bmatrix}, \bar{B} = TB = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \bar{C} = CT^{-1} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}.$$

Therefore, in the new coordinate $z = Tx = [z_1 \;\; z_2 \;\; z_3]^\top$, the non-observable subsystem is as follows:

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ g-K & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}. \tag{1.1}$$

Answers

Moreover, the observable subsystem is as follows:

$$\dot{z}_3 = u.$$

[5 Marks]

d) According to the non-observable subsystem (1.1), the non-observable poles are the eigenvalues of its state matrix, which are $\pm i\sqrt{K-g}$.

[3 Marks]

e) By adding θ_ℓ as a new output, the modified output matrix C is as follows:

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

which yields

$$O = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ K & g-K & 0 \end{bmatrix}$$

which is full rank. Therefore, the new pair (C,A) is observable.

[4 Marks]

Answers

2. Solution

a) According to the block scheme, the state-space representation of the system can be written as follows:

$$\begin{bmatrix} x_1(t+1) \\ x_2(t+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} d_1(t) \\ d_2(t) \end{bmatrix}. \tag{2.1}$$

[3 Marks]

b) The eigenvalues of the state matrix of (2.1) are $\lambda_1 = 1$ and $\lambda_2 = -1$. Accordingly, associated eigenvectors are $v_1 = [1 \;\; 1]^\top$ and $v_2 = [-1 \;\; 1]^\top$, respectively. Therefore, the diagonalizing similarity transformation matrix T can be obtained as $T = [v_1 \;\; v_2]^{-1}$. Therefore, using the similarity transformation matrix T, the system (2.1) can be transformed to

$$\begin{bmatrix} z_1(t+1) \\ z_2(t+1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} z_1(t) \\ z_2(t) \end{bmatrix} + \begin{bmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} d_1(t) \\ d_2(t) \end{bmatrix}. \tag{2.2}$$

[3 Marks]

c) Since $d_1(t)$ and $d_2(t)$ are independent, we can define independent disturbances $\bar{d}_1(t) = \frac{1}{2}d_1(t) + \frac{1}{2}d_2(t)$ and $\bar{d}_2(t) = -\frac{1}{2}d_1(t) + \frac{1}{2}d_2(t)$, such that (2.2) can be restated as follows:

$$\begin{bmatrix} z_1(t+1) \\ z_2(t+1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} z_1(t) \\ z_2(t) \end{bmatrix} + \begin{bmatrix} \bar{d}_1(t) \\ \bar{d}_2(t) \end{bmatrix}. \tag{2.3}$$

The variances of $\bar{d}_1(t)$ and $\bar{d}_2(t)$ are given by

$$V_1 = \text{var}[\bar{d}_1(t)] = \frac{1}{4}\text{var}[d_1(t)] + \frac{1}{4}\text{var}[d_2(t)] = 1$$
$$V_2 = \text{var}[\bar{d}_2(t)] = \frac{1}{4}\text{var}[d_1(t)] + \frac{1}{4}\text{var}[d_2(t)] = 1.$$

The two variables $z_1(t)$ and $z_2(t)$ are decoupled. It is then possible to separately compute the optimal steady-state Kalman predictor for each state component. Let us first consider the variable $z_1(t)$. The general Algebraic Riccati Equation is as follows:

$$P_1 = F_1\Big[P_1 - P_1H_1^\top\big(W_1 + H_1P_1H_1^\top\big)^{-1}H_1P_1\Big]F_1^\top + V_1.$$

By substituting $F_1 = 1$, $H_1 = 1$, $W_1 = \text{var}[\xi_1(t)] = 1$, and $V_1 = 1$, we obtain

$$P_1 = \Big[P_1 - P_1\big(1 + P_1\big)^{-1}P_1\Big] + 1,$$

whose positive solution is $P_1 = (1 + \sqrt{5})/2$. In the same way, the general Algebraic Riccati Equation regarding the variable $z_1(t)$ is as follows:

$$P_2 = F_2\Big[P_2 - P_2H_2^\top\big(W_2 + H_2P_2H_2^\top\big)^{-1}H_2P_2\Big]F_2^\top + V_2.$$

By substituting $F_2 = -1$, $H_2 = 1$, $W_2 = \text{var}[\xi_2(t)] = 2$, and $V_2 = 1$, we obtain

$$P_2 = \Big[P_2 - P_2\big(2 + P_2\big)^{-1}P_2\Big] + 1,$$

whose positive solution is $P_2 = 2$.

[5 Marks]

Answers

d) According to the values of P_1 and P_2 obtained in the answer to Question 2-b), one gets

$$K_1 = F_1 P_1 H_1^\top (W_1 + H_1 P_1 H_1^\top)^{-1} = \frac{1+\sqrt{5}}{2}\left(1+\frac{1+\sqrt{5}}{2}\right)^{-1} = \frac{1+\sqrt{5}}{3+\sqrt{5}}$$
$$K_2 = F_2 P_2 H_2^\top (W_2 + H_2 P_2 H_2^\top)^{-1} = -2(2+2)^{-1} = -\frac{1}{2}.$$

Therefore, the difference equations yielding the one-step ahead optimal steady-state Kalman prediction are as follows:

$$\begin{aligned} \hat{z}_1(t+1|t) &= \hat{z}_1(t|t-1) + \frac{1+\sqrt{5}}{3+\sqrt{5}}(y_1(t) - \hat{z}_1(t|t-1)), \\ \hat{z}_2(t+1|t) &= -\hat{z}_2(t|t-1) - \frac{1}{2}(y_2(t) - \hat{z}_2(t|t-1)). \end{aligned} \tag{2.4}$$

[4 Marks]

e) According to (2.3) and (2.4), by defining $\tilde{z}_i(t) = z_i(t) - \hat{z}_i(t|t-1), i = 1,2$, one gets

$$\tilde{z}_i(t+1) = (F_i - K_i)\tilde{z}_i(t) + \bar{d}_i(t) - K_i\xi_i(t), \tag{2.5}$$

in which $F_1 = 1$, $F_2 = -1$, $K_1 = (1+\sqrt{5})/(3+\sqrt{5})$, and $K_2 = -1/2$. Since both the systems given in (2.5) are stable, they are stationary. Moreover, they are independent. Therefore,

$$\mathrm{cov}[z(t) - \hat{z}(t|t-1)] = \begin{bmatrix} \mathrm{var}[\tilde{z}_1] & 0 \\ 0 & \mathrm{var}[\tilde{z}_1] \end{bmatrix} = \begin{bmatrix} \frac{\mathrm{var}[\bar{d}_1(t)] + K_1^2 \mathrm{var}[\xi_1(t)]}{1-(F_1-K_1)^2} & 0 \\ 0 & \frac{\mathrm{var}[\bar{d}_2(t)] + K_2^2 \mathrm{var}[\xi_2(t)]}{1-(F_2-K_2)^2} \end{bmatrix}$$

which is

$$\mathrm{cov}[z(t) - \hat{z}(t|t-1)] = \begin{bmatrix} 1.6180 & 0 \\ 0 & 1.6667 \end{bmatrix}.$$

[5 Marks]

Answers

3. Solution

a) According to the state-space representation of the system, the state matrix and the output matrix of the system are

$$A = \begin{bmatrix} 3-\alpha & 1+\alpha \\ 2 & \alpha \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Thus, the observability matrix of the pair (C,A) is as follows:

$$O = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3-\alpha & 1+\alpha \end{bmatrix}.$$

Therefore, O is full rank for all real α except $\alpha = -1$. Then, the system is observable for all real α except $\alpha = -1$.

[3 Marks]

b) For $\alpha = 1$, we have

$$\begin{aligned} \dot{x} &= Ax + Bu, \\ y &= Cx \end{aligned}$$

where

$$A = \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Now, a full-order state observer can be designed as follows:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}).$$

Since $e(t) = x(t) - \hat{x}(t)$, one gets

$$\dot{e} = (A - LC)e = Fe. \tag{3.1}$$

By letting $L = [\ell_1 \ \ \ell_2]^\top$,

$$F = \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} \ell_1 \\ \ell_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 2-\ell_1 & 2 \\ 2-\ell_2 & 1 \end{bmatrix}.$$

The eigenvalues of F can be obtained from the solution of the following equation:

$$|\lambda I_2 - F| = \lambda^2 - (2 - \ell_1 + 1)\lambda + 2 - \ell_1 - 2(2 - \ell_2) = 0.$$

Therefore, to set the eigenvalues of F at -1 and -2, we set $\ell_1 = 6$ and $\ell_2 = 5$.

[4 Marks]

c) According to the state-space representation of the system, the state matrix and the output matrix of the system when $\alpha = -1$ are as

$$A = \begin{bmatrix} 4 & 0 \\ 2 & -1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Thus, the observability matrix of the pair (C,A) is as follows:

$$O = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 0 \end{bmatrix}$$

Answers

from which we deduce that the system is not observable, and that the dimension of the non-observable subspace is one. From the expression of A and C, it follows that the non-observable eigenvalue is -1. Therefore, it is possible to design L such that the eigenvalues of $F = A - LC$ are -1 and -2.

[3 Marks]

d) By proceeding in the same way as in the answer to Question 3-b), we obtain

$$\dot{e}(t) = Fe(t) + Bf.$$

Therefore, we have

$$\varepsilon(t) = Ce(t) = C\left(\mathrm{e}^{Ft}e(0) + \int_0^t \mathrm{e}^{F(t-\tau)}Bf d\tau\right).$$

To obtain e^{Ft}, we first diagonalize F by using a similarity transformation matrix T, where T^{-1} has as columns right eigenvectors of F as follows:

$$T^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}$$

and accordingly

$$T = \begin{bmatrix} -1 & 1 \\ 3 & -2 \end{bmatrix}.$$

By defining $z = Te$, we obtain

$$\dot{z} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} z$$

whose solution from $t = 0$ is

$$\begin{aligned} z(t) &= \begin{bmatrix} \mathrm{e}^{-t} & 0 \\ 0 & \mathrm{e}^{-2t} \end{bmatrix} Te(0) + \int_0^t \begin{bmatrix} \mathrm{e}^{-(t-\tau)} & 0 \\ 0 & \mathrm{e}^{-2(t-\tau)} \end{bmatrix} TBf d\tau \\ &= \begin{bmatrix} -\mathrm{e}^{-t} & \mathrm{e}^{-t} \\ 3\mathrm{e}^{-2t} & -2\mathrm{e}^{-2t} \end{bmatrix} e(0) + \int_0^t \begin{bmatrix} 0 \\ \mathrm{e}^{-2(t-\tau)}f \end{bmatrix} d\tau \\ &= \begin{bmatrix} -\mathrm{e}^{-t} & \mathrm{e}^{-t} \\ 3\mathrm{e}^{-2t} & -2\mathrm{e}^{-2t} \end{bmatrix} e(0) + \begin{bmatrix} 0 \\ \frac{1}{2}(1-\mathrm{e}^{-2t}) \end{bmatrix} f. \end{aligned}$$

Therefore,

$$\begin{aligned} \varepsilon(t) &= Ce(t) = CT^{-1}z(t) \\ &= \begin{bmatrix} -2\mathrm{e}^{-t} + 3\mathrm{e}^{-2t} & 2\mathrm{e}^{-t} - 2\mathrm{e}^{-2t} \end{bmatrix} e(0) + \frac{1}{2}(1-\mathrm{e}^{-2t})f. \end{aligned} \tag{3.2}$$

[3 Marks]

e) According to (3.2), at $t = 1$, we have

$$\varepsilon(1) - \begin{bmatrix} -2\mathrm{e}^{-1} + 3\mathrm{e}^{-2} & 2\mathrm{e}^{-1} - 2\mathrm{e}^{-2} \end{bmatrix} e(0) = \frac{1}{2}(1-\mathrm{e}^{-2})f$$

implying that

$$|\varepsilon(1)| + \left\| \begin{bmatrix} -2\mathrm{e}^{-1} + 3\mathrm{e}^{-2} & 2\mathrm{e}^{-1} - 2\mathrm{e}^{-2} \end{bmatrix} \right\| \|e(0)\| \geq \frac{1}{2}(1-\mathrm{e}^{-2})\|f\|.$$

Answers

Therefore, if

$$|f| > \frac{2}{1-\mathrm{e}^{-2}}\left(\left\|\begin{bmatrix}-2\mathrm{e}^{-1}+3\mathrm{e}^{-2} & 2\mathrm{e}^{-1}-2\mathrm{e}^{-2}\end{bmatrix}\right\| \|e(0)\|+1\right)$$
$$= 2.3130(0.5701+1) = 3.6316,$$

then, $|\varepsilon(1)| > 1$.

[3 Marks]

f) By considering f as a new state, we have $\dot{f} = 0$. In this case, by defining the state vector $\bar{x} = [x^\top \quad f]^\top$, the state-space representation of the system can be stated as

$$\dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u,$$
$$\bar{y} = \bar{C}\bar{x}$$

where

$$\bar{A} = \begin{bmatrix}2 & 2 & 1\\ 2 & 1 & 1\\ 0 & 0 & 0\end{bmatrix}, \bar{C} = \begin{bmatrix}1 & 0 & 0\end{bmatrix}, \bar{B} = \begin{bmatrix}1\\ 1\\ 0\end{bmatrix}.$$

Now, the observability matrix of the pair $(\bar{C}, \bar{A})$ is as follows:

$$O = \begin{bmatrix}\bar{C}\\ \bar{C}\bar{A}\\ \bar{C}\bar{A}^2\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 2 & 2 & 1\\ 8 & 6 & 4\end{bmatrix}.$$

Therefore, O is full rank and the fault is observable.

[4 Marks]

Answers

4. Solution

a) We have

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & 2 \\ 0 & -1 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & c \end{bmatrix}.$$

Thus, the observability matrix of the pair (C,A) is as follows:

$$O = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & c \\ 0 & -1-c & c \\ 0 & -1-2c & -2-c \end{bmatrix}.$$

Since $\det(O) = 3c^2 + 4c + 2$ has no real roots, for all real c, the system is observable.

[4 Marks]

b) Since the rank of C is 1, the order of the reduced order state observer is $n - p = 3 - 1 = 2$.

[3 Marks]

c) The reduced-order state observer has the following form:

$$\begin{aligned} \dot{\xi} &= F\xi + Hy, \\ \hat{x}_r &= \xi + Ly \end{aligned} \tag{4.1}$$

where F and H are defined as

$$\begin{aligned} F &= A_{22} - LA_{12}, \\ H &= A_{21} - LA_{11} + FL, \end{aligned} \tag{4.2}$$

in which

$$\begin{aligned} A_{11} &= 0 \\ A_{12} &= \begin{bmatrix} -1 & 0 \end{bmatrix} \\ A_{21} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ A_{22} &= \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \end{aligned} \tag{4.3}$$

and $L = [\ell_1 \ \ \ell_2]^\top$ is designed to assign the eigenvalues of F. We have

$$F = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} \ell_1 \\ \ell_2 \end{bmatrix} \begin{bmatrix} -1 & 0 \end{bmatrix} = \begin{bmatrix} 1+\ell_1 & 2 \\ -1+\ell_2 & 1 \end{bmatrix}.$$

Now, by setting $\ell_1 = -4$ and $\ell_2 = -1$, the eigenvalues of F are set as -1 and -1.

[7 Marks]

d) The state $x_1(t)$ can be obtained as

$$x_1(t) = \int_0^t f d\tau - \int_0^t x_2(\tau) d\tau = ft - \int_0^t x_2(\tau) d\tau.$$

Answers

Since the subsystem (x_2, x_3) is asymptotically stable, x_2 converges exponentially to zero, and thus $\int_0^t x_2(\tau)d\tau$ is bounded. Thus, $x_1(t)$ is unbounded as $t \to \infty$.

[6 Marks]

Answers